

Wandering Black Hole Feeds on Its Own Gravitational Wake

A 35,000-solar-mass intermediate-mass black hole accretes from a dense downstream wake, revealing a new growth channel.

arXiv:2608.10719 — Wandering 35,000-solar-mass black hole fed by gravitational wake

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Background

Intermediate-mass black holes are considered seeds of supermassive black holes, but many remain displaced from galactic nuclei. Off-nuclear black holes lack access to conventional fueling channels, leaving their fuel supply uncertain.

Direct observational evidence for gravitational-wake accretion onto wandering black holes has remained elusive.

What this paper is selling

This paper reports a wandering 35,000-solar-mass black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals distinct upstream and downstream gas components and rapid broad-line variability.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm black hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense wake, optically thick absorbers.
- Optically thick absorbers within capture radius drive rapidly changing-look broad-line variability.

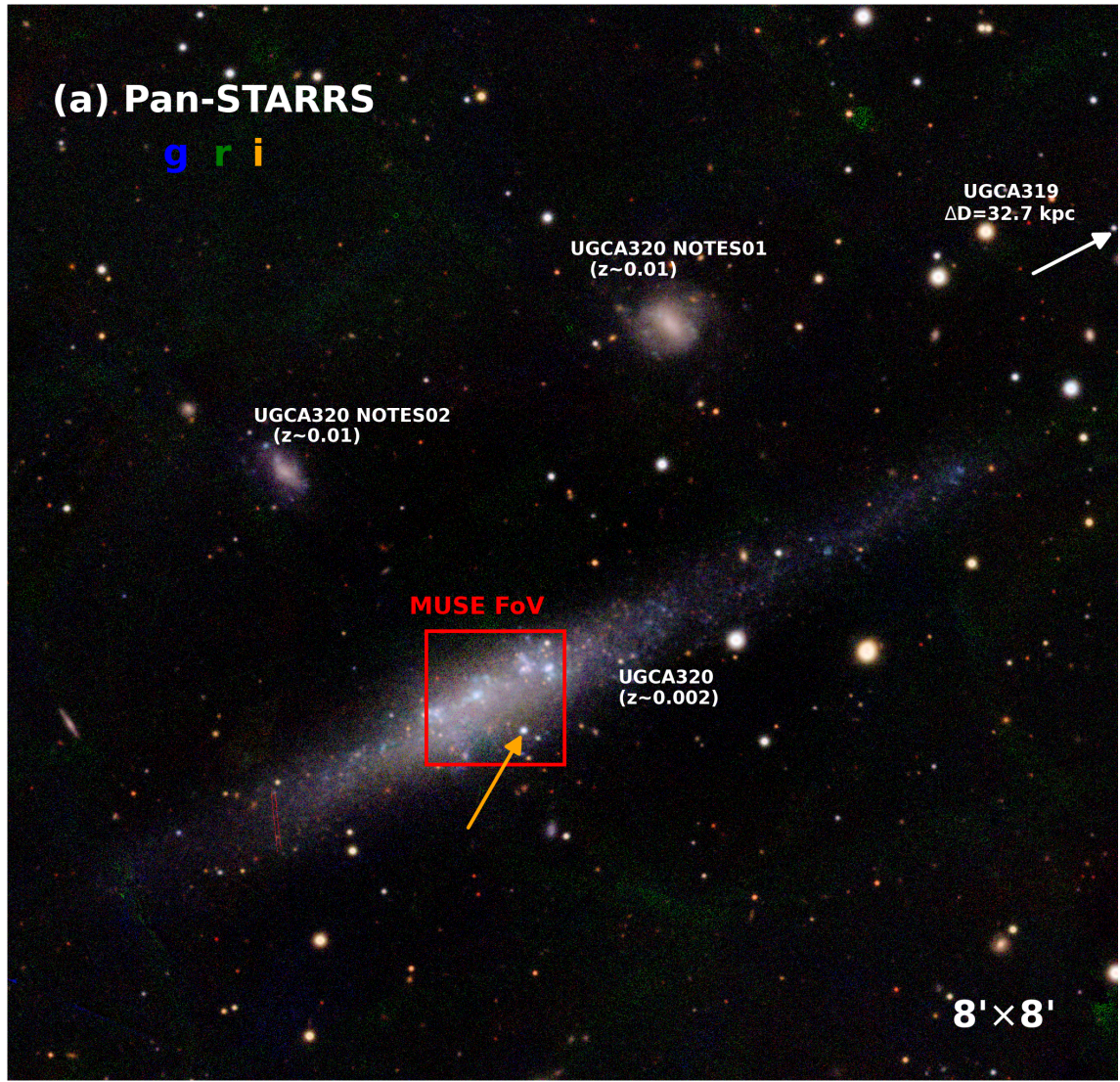


Figure 1 | HST and MUSE views pinpoint the wandering IMBH in UGCA 320, with broad H α and [O III] contours marking its location.

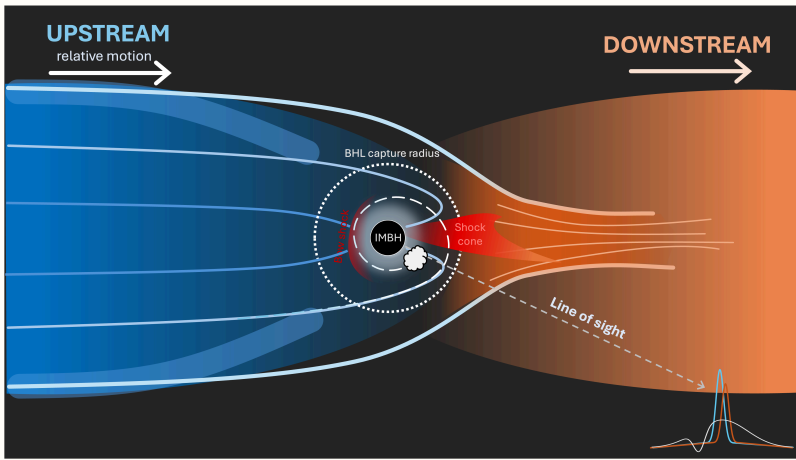


Figure 4 | Bondi–Hoyle–Lyttleton accretion schematic shows upstream gas deflected into a dense downstream wake that feeds the black hole.